

Improved irrationality-exponent bounds for $\log 3$ and $\log_2 3$

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Abstract

We prove that, for all sufficiently large $H = \max(|h_1|, |h_2|)$ and all integers h_0, h_1, h_2 with $(h_1, h_2) \neq (0, 0)$,

$$|h_0 + h_1 \log 2 + h_2 \log 3| > H^{-4.104104}.$$

Consequently, $\mu(\log 3) \leq 5.104104$ and $\mu(\log 3 / \log 2) \leq 5.104104$, improving the previously published bound 5.116201. The proof constructs two positive integer linear forms with a common coefficient in $\log(4/3)$ and $\log(9/8)$ from a product of twenty-one symmetrized integer polynomials with exact rational weights. Coefficient-content identities provide integrality, the Laplace principle gives two distinct decay rates, and Cauchy's estimate controls the common coefficient. The finite inequalities are certified in exact rational arithmetic by real-root isolation, Bernstein-basis sign certificates, and rational logarithm enclosures. A limit-superior two-form criterion then yields the homogeneous estimate and both irrationality-exponent bounds.

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1. Introduction and statement of the result

Throughout, $\log$ denotes the natural logarithm. For an irrational real number ξ , its irrationality exponent is

$$\mu(\xi) = \inf \left\{ \mu > 0 : 0 < |\xi - p/q| < q^{-\mu} \text{ has only finitely many solutions } (p, q) \in \mathbb{Z} \times \mathbb{Z}_{\geq 1} \right\}.$$

Dirichlet's theorem gives $\mu(\xi) \geq 2$ for every irrational ξ . Obtaining an explicit upper bound for a particular transcendental constant is a different problem:

one must construct rational approximations and control, simultaneously and effectively, both their denominators and their errors.

The two numbers considered here are connected by the linear form

$$q \log 3 - p \log 2 = \log \frac{3^q}{2^p}.$$

Thus rational approximation to $\log 3 / \log 2 = \log_2 3$ measures how closely powers of 2 and 3 can approach one another on a logarithmic scale. A lower bound for arbitrary integer forms in $1, \log 2, \log 3$ is stronger than separate irrationality-exponent estimates and yields both of them by simple specialization. Such bounds also have applications outside number theory: Draper and Saad use the exponent 5.116201 in a space-entropy lower bound for exact sampling from the Bernoulli distribution with parameter $1/3$ [1].

Effective bounds for logarithms of rational numbers are commonly obtained from definite integrals whose partial-fraction expansions are rational linear forms in logarithms; see Zudilin's survey [2]. The quality of such a construction is governed by three competing quantities: the exponential decay of the integral, the growth of the logarithm coefficient, and the arithmetic cost of clearing denominators. Rhin's simultaneous construction gave $\mu(\gamma) < 8.616$ for every nonzero $\gamma \in \mathbb{Q} \log 2 + \mathbb{Q} \log 3$ [3]. For $\log 3$, Salikhov obtained $\mu(\log 3) \leq 5.125 \dots$ [4], and Wu and Wang reduced the bound to 5.1163051 using an arithmetic optimization with symmetrized linear factors [5]. Bondareva, Luchin, and Salikhov added a quadratic symmetrized factor and proved the homogeneous exponent 4.116201 [6]. In particular, their result gives

$$\mu(\log 3) \leq 5.116201, \quad \mu(\log 3 / \log 2) \leq 5.116201. \quad (1)$$

The present construction enlarges the symmetrized factor family to twenty-one polynomials of degrees between one and five and uses an exactly defined positive rational weight vector. The local coefficient-content balances are retained, while the resulting coefficient-growth-to-decay quotient is reduced from 4.116201 to below 4.104104. This gives the following strict improvement of (1).

Theorem 1.1. *Let $h_0, h_1, h_2 \in \mathbb{Z}$, with $(h_1, h_2) \neq (0, 0)$, and set $H = \max(|h_1|, |h_2|)$. For all sufficiently large H ,*

$$|h_0 + h_1 \log 2 + h_2 \log 3| > H^{-4.104104}.$$

Corollary 1.2. *One has*

$$\mu(\log 3) \leq 5.104104, \quad \mu\left(\frac{\log 3}{\log 2}\right) \leq 5.104104.$$

Here is the structure of the proof. With

$$\theta_1 = \log(4/3), \quad \theta_2 = \log(9/8),$$

the symmetrized rational function produces two positive integrals over $[35, 40]$ and $[40, 42]$. After an exact coefficient-content calculation and partial fractions, multiplication by a common clearing factor turns them into integer linear forms

$$q_n \theta_1 - p_{1,n}, \quad q_n \theta_2 - p_{2,n}$$

with the same coefficient q_n . Positivity and the Laplace principle reduce their exponential decay to maxima of a single real function on two adjacent intervals. Cauchy's coefficient estimate reduces the growth of q_n to a maximum on a complex circle. Exact root isolation, Bernstein-basis signs, and rational logarithm enclosures then prove that the two decay rates are distinct and that the quotient of coefficient growth by the smaller decay rate is less than

$$4.104103101316511697013643 < 4.104104.$$

Finally, the unimodular relations

$$\log(4/3) = 2 \log 2 - \log 3, \quad \log(9/8) = -3 \log 2 + 2 \log 3$$

transfer the resulting homogeneous estimate to $\log 2$ and $\log 3$.

Two features require explicit treatment. First, the Cauchy argument gives a limit superior for the common coefficient rather than a full exponential limit. Section 2 therefore proves the precise limit-superior version of Hata's two-form criterion [7, Lemma 2.1] needed here. Second, the optimization inequalities are too large for useful printed expansion but form a finite certificate over $\mathbb{Q}$. The main proof states exactly the certified polynomial and interval assertions it uses; their exact reproduction is described in Appendix B, and the complete certificate accompanies the paper as supplementary material. No floating-point comparison is used in a theorem-facing decision.

Section 2 establishes the Diophantine criterion. Sections 3 and 4 define the construction and prove integrality. Sections 5–7 determine the two decay rates and bound the common coefficient, and Section 8 completes the proof. The factor data are listed in Appendix A, while the finite certificate and its reproduction are recorded in Appendix B.

2. A two-linear-form criterion

The next lemma is a limit-superior variant of Hata's two-form criterion [7, Lemma 2.1]. We include the proof because replacing a limit by a limit superior is not merely notational: possible cancellation between the two small forms must be controlled uniformly in their integer coefficients.

Lemma 2.1 (two-form criterion). *Let $\theta_1, \theta_2 \in \mathbb{R}$. Suppose that integers $q_n, p_{1,n}, p_{2,n}$ satisfy*

$$\varepsilon_{j,n} = q_n \theta_j - p_{j,n} \neq 0 \quad (j = 1, 2)$$

for all sufficiently large n , and suppose that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log |\varepsilon_{j,n}| = -\tau_j \quad (j = 1, 2), \quad \limsup_{n \rightarrow \infty} \frac{1}{n} \log |q_n| \leq \lambda, \quad (2)$$

where $\tau_1, \tau_2 > 0$, $\tau_1 \neq \tau_2$, and $\lambda \geq 0$. Put $\tau = \min(\tau_1, \tau_2)$. For every $\nu > \lambda/\tau$ there is $H_0(\nu)$ such that

$$|h_0 + h_1 \theta_1 + h_2 \theta_2| > H^{-\nu}$$

whenever $h_0, h_1, h_2 \in \mathbb{Z}$, $H = \max(|h_1|, |h_2|) \geq H_0(\nu)$, and $(h_1, h_2) \neq (0, 0)$.

Proof. Interchanging the two coordinates if necessary, assume $0 < \tau_1 < \tau_2$. Write $d = \tau_2 - \tau_1$ and $T = \tau_2$. Fix $\nu > \lambda/\tau_1$. Choose $c > 0$ and then $\eta > 0$ so small that

$$dc > 2\eta(2 + 3c), \quad c\tau_1 > \eta(2 + 2c), \quad (3)$$

$$\frac{\Gamma}{\tau_1 - \eta} < \nu, \quad (4)$$

$$\Gamma = (\lambda + \eta)(1 + 2c) + 2cT + (2 + 2c)\eta. \quad (5)$$

Such a choice is possible: take $c \downarrow 0$ and then $\eta = o(c)$; the quotient in (4) tends to λ/τ_1 .

After increasing N , the hypotheses give, for $n \geq N$ and $j = 1, 2$,

$$e^{-(\tau_j + \eta)n} \leq |\varepsilon_{j,n}| \leq e^{-(\tau_j - \eta)n}, \quad 0 < |q_n| \leq e^{(\lambda + \eta)n}. \quad (6)$$

The eventual nonvanishing of q_n follows as well: if $q_n = 0$ when both errors have absolute value less than one, then $\varepsilon_{j,n} = -p_{j,n}$ cannot be a nonzero integer.

Fix (h_0, h_1, h_2) and put

$$S_m = |h_1| |\varepsilon_{1,m}| + |h_2| |\varepsilon_{2,m}|, \quad E_m = h_1 \varepsilon_{1,m} + h_2 \varepsilon_{2,m}.$$

Let $n \geq N + 1$ be the least integer for which $S_n < 1/2$. This integer exists. Since one of $|h_1|$ and $|h_2|$ equals H ,

$$S_m \geq H \min(|\varepsilon_{1,m}|, |\varepsilon_{2,m}|).$$

For every fixed M , the nonzero errors at the finitely many indices $N+1, \dots, M$ therefore show that $S_m \geq 1/2$ throughout that range once H is sufficiently large. Thus $n \rightarrow \infty$ uniformly with H ; in particular, $n > N + 1$ and $S_{n-1} \geq 1/2$ for sufficiently large H . By (6),

$$\frac{1}{2} \leq S_{n-1} \leq 2H e^{-(\tau_1 - \eta)(n-1)},$$

so

$$n \leq 1 + \frac{\log(4H)}{\tau_1 - \eta}. \quad (7)$$

Set

$$k_1 = \lceil (1 + c)n \rceil, \quad k_2 = \lceil (1 + 2c)n \rceil.$$

The inequality $S_n < 1/2$ and the lower bounds in (6) imply

$$|h_j| < \frac{1}{2} e^{(\tau_j + \eta)n} \quad (j = 1, 2).$$

Consequently, for $k \in \{k_1, k_2\}$,

$$|h_j \varepsilon_{j,k}| \leq \frac{1}{2} \exp(-\tau_j(k - n) + \eta(n + k)).$$

The second inequality in (3) now shows that $S_{k_1}, S_{k_2} < 1/2$ once H , and hence n , is sufficiently large.

Suppose first that $h_1 h_2 \neq 0$, and define

$$R_k = \frac{|h_1 \varepsilon_{1,k}|}{|h_2 \varepsilon_{2,k}|}.$$

The coefficient quotient cancels from R_{k_2}/R_{k_1} . From (6),

$$\frac{R_{k_2}}{R_{k_1}} \geq \exp((d - 2\eta)k_2 - (d + 2\eta)k_1) > 4$$

for all sufficiently large n , by the first inequality in (3). Hence at least one $k \in \{k_1, k_2\}$ satisfies $R_k \notin [1/2, 2]$. Choose such a k . If one of h_1, h_2 is zero, choose $k = k_1$ instead.

Since $S_{n-1} \geq 1/2$, there is an index $j_0 \in \{1, 2\}$ with

$$|h_{j_0} \varepsilon_{j_0, n-1}| \geq \frac{1}{4}.$$

Using first the upper and then the lower error bounds in (6), and using $k \leq (1 + 2c)n + 1$, gives

$$\begin{aligned} |h_{j_0} \varepsilon_{j_0, k}| &\geq \frac{1}{4} \exp((\tau_{j_0} - \eta)(n - 1) - (\tau_{j_0} + \eta)k) \\ &\geq \frac{e^{-2T}}{4} \exp(-(2cT + (2 + 2c)\eta)n). \end{aligned}$$

If both coefficients are nonzero, the choice $R_k \notin [1/2, 2]$ implies that one of the two summands in E_k has more than twice the absolute value of the other. If one coefficient is zero, there is only one summand. In either case,

$$|E_k| \geq \frac{e^{-2T}}{8} \exp(-(2cT + (2 + 2c)\eta)n). \quad (8)$$

Put

$$\Lambda = h_0 + h_1 \theta_1 + h_2 \theta_2, \quad A_k = q_k h_0 + p_{1,k} h_1 + p_{2,k} h_2 \in \mathbb{Z}.$$

Then $q_k \Lambda = A_k + E_k$ and $|E_k| \leq S_k < 1/2$. If $A_k \neq 0$, then $|\Lambda| > 1/(2|q_k|)$. If $A_k = 0$, then $|\Lambda| = |E_k|/|q_k|$, and (8) applies. Together with (6), both cases give a constant $C > 0$, independent of the coefficients, such that

$$|\Lambda| \geq C e^{-\Gamma n}.$$

Using (7), this is at least a positive constant times $H^{-\Gamma/(\tau_1 - \eta)}$. By (4), it exceeds $H^{-\nu}$ for all sufficiently large H . This proves the lemma. $\square$

3. The symmetrized integral construction

The numerator will be invariant under $x \mapsto 70 - x$. It is therefore convenient to put

$$t = (x - 35)^2.$$

The twenty-one integer polynomials $P_k(t)$ are listed, together with their local coefficient profiles, in Table A.1 of Appendix A. Their degrees are between one and five.

For $p \in \{2, 3, 5, 7\}$, define

$$\tau_p(P_k) = \min_j v_p([y^j]P_k((840y - 35)^2)), \quad (9)$$

where zero coefficients are omitted from the minimum. Thus $p^{\tau_p(P_k)}$ is the exact p -part of the coefficient content relevant to the scaling used below.

3.1. Exact weights

Set

$$z_0 = 26 \frac{-899 + 60i}{901}, \quad t_0 = (z_0 - 35)^2 = \frac{3012564681 - 171316080i}{811801}. \quad (10)$$

In particular, $|z_0| = 26$. Direct exact substitution gives $P_k(t_0) \neq 0$ for every k . For $1 \leq k \leq 21$, put

$$\ell_k = 2(z_0 - 35) \frac{P'_k(t_0)}{P_k(t_0)} \in \mathbb{Q}(i). \quad (11)$$

The weights $\alpha_k \in \mathbb{Q}_{>0}$ are defined exactly as follows. For $7 \leq k \leq 21$, set $\alpha_k = u_k/10^{12}$, where

k	7	8	9	10
u_k	119873656	878336597	101691070	103130300
k	11	12	13	14
u_k	219953072	757245952	781996563	232727309
k	15	16	17	18
u_k	630848726	330420926	232427174	314945936
k	19	20	21	
u_k	103903210	14385085	9664407	

The first six weights are the unique rational solution of

$$\sum_{k=1}^{21} \alpha_k \deg P_k = \frac{3}{2}, \quad (12)$$

$$\sum_{k=1}^{21} \alpha_k \tau_7(P_k) = 2, \quad \sum_{k=1}^{21} \alpha_k \tau_5(P_k) = 2, \quad (13)$$

$$\sum_{k=1}^{21} \alpha_k \tau_3(P_k) = 1, \quad (14)$$

$$\sum_{k=1}^{21} \alpha_k \ell_k = \frac{1}{z_0} - \frac{1}{70 - z_0}. \quad (15)$$

The last line means equality of real and imaginary parts, so the system has six rational equations. Exact Gaussian elimination shows that the resulting 6×6 coefficient matrix is nonsingular and that all six weights are positive. For orientation,

k	α_k	k	α_k	k	α_k
1	0.492767795602391	8	0.000878336597000	15	0.000630848726000
2	0.497597698986041	9	0.000101691070000	16	0.000330420926000
3	0.484860590979781	10	0.000103130300000	17	0.000232427174000
4	0.000043741872954	11	0.000219953072000	18	0.000314945936000
5	0.000129471059917	12	0.000757245952000	19	0.000103903210000
6	0.000176164327825	13	0.000781996563000	20	0.000014385085000
7	0.000119873656000	14	0.000232727309000	21	0.000009664407000

Let D be the least common multiple of the denominators of the α_k , and write

$$W_k = D\alpha_k \in \mathbb{Z}_{>0}.$$

The integer D has 1061 decimal digits. Its exact value and all W_k are part of the supplementary certificate. We let n tend to infinity through the positive multiples of $2D$. Then every $\alpha_k n$ is an even integer.

The numerical search that led to these parameters is not part of the proof. Its provenance and the subsequent exact rationalization are recorded in Appendix B and in the supplementary material. No optimality is claimed for the factor family.

4. Arithmetic and integer linear forms

Put

$$A_n(x) = \prod_{k=1}^{21} P_k((x-35)^2)^{\alpha_k n}, \quad R_n(x) = \frac{A_n(x)}{x^{n+1}(70-x)^{n+1}}. \quad (16)$$

The degree balance gives $\deg_x A_n = 3n$. The numerator has integer coefficients, and all of its factor exponents are even. Consequently R_n is nonnegative on the two real intervals used below.

4.1. The scaled numerator

Set

$$s_2 = \sum_{k=1}^{21} \alpha_k \tau_2(P_k), \quad a = 4 - s_2. \quad (17)$$

The exact weights give

$$a = 0.43339168265643340846553249867665 \dots > 0. \quad (18)$$

Because n is a multiple of $2D$, an is a nonnegative integer.

Lemma 4.1 (integer scaled numerator). *For every admissible n ,*

$$B_n(y) = \frac{2^{an} A_n(840y)}{58800^n} \quad (19)$$

belongs to $\mathbb{Z}[y]$.

Proof. For each k , definition (9) says that every coefficient of $P_k((840y-35)^2)$ is divisible by

$$2^{\tau_2(P_k)} 3^{\tau_3(P_k)} 5^{\tau_5(P_k)} 7^{\tau_7(P_k)}.$$

After raising the factors to the integer powers $\alpha_k n$ and multiplying, (13)–(14) supply respectively 7^{2n} , 5^{2n} , and 3^n . Definition (17) supplies 2^{4n} after multiplication by 2^{an} . Since

$$58800 = 2^4 \cdot 3 \cdot 5^2 \cdot 7^2,$$

the quotient in (19) has integer coefficients. □

4.2. Partial fractions and denominator clearing

Symmetry under $x \mapsto 70 - x$ gives

$$R_n(x) = Q_{n-2}(x) + \sum_{i=1}^{n+1} a_{i,n} \left(\frac{1}{x^i} + \frac{1}{(70-x)^i} \right), \quad (20)$$

where $Q_{n-2} \in \mathbb{Z}[x]$. Indeed, the polynomial denominator in (16) has leading coefficient ± 1 , so polynomial division preserves integrality, and symmetry identifies the two principal parts.

Lemma 4.2 (partial-fraction denominators). *For $1 \leq i \leq n+1$ there is an integer $M_{i,n}$ such that*

$$a_{i,n} = \frac{2^{-an} 840^{i-1}}{70} M_{i,n}. \quad (21)$$

Proof. At $x = 0$,

$$a_{i,n} = [x^{n+1-i}] \frac{A_n(x)}{(70-x)^{n+1}}.$$

Put $j = n+1-i$ and substitute $x = 840y$. By Lemma 4.1,

$$\begin{aligned} a_{i,n} &= 840^{-j} \frac{2^{-an} 58800^n}{70^{n+1}} [y^j] B_n(y) (1-12y)^{-n-1} \\ &= \frac{2^{-an} 840^{n-j}}{70} [y^j] B_n(y) (1-12y)^{-n-1}. \end{aligned}$$

Now $n-j = i-1$ and $(1-12y)^{-n-1} \in \mathbb{Z}[[y]]$. □

Let

$$\Delta_n = \text{lcm}(1, 2, \dots, n), \quad C_n = 70 \cdot 2^{an} \Delta_n, \quad (22)$$

and define

$$I_{1,n} = \int_{35}^{40} R_n(x) dx, \quad I_{2,n} = \int_{40}^{42} R_n(x) dx. \quad (23)$$

Proposition 4.3 (two positive integer forms). *There are integers $q_n, p_{1,n}, p_{2,n}$ such that*

$$C_n I_{1,n} = q_n \log \frac{4}{3} - p_{1,n}, \quad (24)$$

$$C_n I_{2,n} = q_n \log \frac{9}{8} - p_{2,n}, \quad (25)$$

where the common coefficient is $q_n = C_n a_{1,n} \in \mathbb{Z}$. Both left-hand sides are strictly positive.

Proof. For integers $0 < u < v < 70$,

$$\int_u^v \left(\frac{1}{x} + \frac{1}{70-x} \right) dx = \log \frac{v(70-u)}{u(70-v)}, \quad (26)$$

$$\int_u^v \left(\frac{1}{x^i} + \frac{1}{(70-x)^i} \right) dx = \frac{u^{1-i} - v^{1-i} + (70-v)^{1-i} - (70-u)^{1-i}}{i-1} \quad (27)$$

for $i \geq 2$. The logarithm in (26) is $\log(4/3)$ for $(u, v) = (35, 40)$ and $\log(9/8)$ for $(u, v) = (40, 42)$.

Every endpoint in (27) belongs to $\{28, 30, 35, 40, 42\}$ and divides 840. By (21), multiplication by $70 \cdot 2^{an}$ clears its $(i-1)$ st power, while Δ_n clears the factor $i-1$. The polynomial part has integer coefficients and degree $n-2$, so Δ_n also clears all denominators in its integral between integer endpoints. This proves (24)–(25). Positivity follows because R_n is nonnegative and not identically zero on either interval. $\square$

Since $\log \Delta_n = \psi(n)$, the prime number theorem in the form $\psi(n) \sim n$ gives

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \Delta_n = 1$$

[8, Ch. 6]. Hence, along the admissible values of n ,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log C_n = c := 1 + a \log 2. \quad (28)$$

5. Exact logarithm enclosures

Lemma 5.1. *For $0 \leq z < 1$ and $N \geq 1$,*

$$2 \sum_{j=0}^{N-1} \frac{z^{2j+1}}{2j+1} \leq \log \frac{1+z}{1-z} \leq 2 \sum_{j=0}^{N-1} \frac{z^{2j+1}}{2j+1} + \frac{2z^{2N+1}}{(2N+1)(1-z^2)}.$$

Consequently every logarithm of a positive rational number can be enclosed between explicit rational numbers.

Proof. Use the positive series $\operatorname{atanh} z = \sum_{j \geq 0} z^{2j+1}/(2j+1)$ and bound every omitted denominator below by $2N+1$. For a positive rational y , choose the exact integer k with $2^k \leq y < 2^{k+1}$ and write

$$\log y = k \log 2 + \log \frac{1+u}{1-u}, \quad u = \frac{y/2^k - 1}{y/2^k + 1}, \quad 0 \leq u < \frac{1}{3}.$$

The same series with $z = 1/3$ encloses $\log 2 = \log((1+z)/(1-z))$, so every term in the displayed decomposition has explicit rational lower and upper bounds. $\square$

6. The two real decay rates

For $0 \leq t \leq 49$, define

$$g(t) = \frac{\prod_{k=1}^{21} |P_k(t)|^{\alpha_k}}{1225 - t}. \quad (29)$$

This is a continuous nonnegative function. On the two integration intervals,

$$R_n(x) = \frac{g((x-35)^2)^n}{1225 - (x-35)^2}. \quad (30)$$

Put

$$M_1 = \log \max_{0 \leq t \leq 25} g(t), \quad M_2 = \log \max_{25 \leq t \leq 49} g(t). \quad (31)$$

Lemma 6.1 (elementary Laplace principle). *Let f be continuous and non-negative, but not identically zero, on a nondegenerate compact interval K , and let w be continuous and strictly positive there. Then*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \int_K f(x)^n w(x) dx = \log \max_K f.$$

Proof. The upper bound follows from $\int_K f^n w \leq (\max_K f)^n \int_K w$. For the lower bound, choose a point at which f attains its positive maximum. For every $\delta > 0$, the function f is at least $\max_K f - \delta$ on a one-sided or two-sided neighborhood of that point, and w has a positive lower bound there. Let $n \rightarrow \infty$ and then $\delta \downarrow 0$. $\square$

Away from the zeros of the P_k ,

$$\frac{d}{dt} \log g(t) = \sum_{k=1}^{21} \alpha_k \frac{P'_k(t)}{P_k(t)} + \frac{1}{1225 - t}.$$

Let $P(t) = \prod_{k=1}^{21} P_k(t)$, and let $H(t)$ be the primitive part, with positive content removed, of

$$(1225 - t) \sum_{k=1}^{21} W_k P'_k(t) \frac{P(t)}{P_k(t)} + DP(t). \quad (32)$$

It is an integer polynomial of degree 85.

Lemma 6.2 (real critical points). *The polynomial H is squarefree and coprime to P . It has exactly 63 roots in $[0, 49]$: 41 in $(0, 25)$ and 22 in $(25, 49)$. Exact rational root isolation and outward rational logarithm bounds give*

$$-2.722957495668415052959634 < M_1 < -2.722957495668409711051756, \quad (33)$$

$$-2.722977562389619550059089 < M_2 < -2.722977562389619502813003. \quad (34)$$

In particular,

$$M_1 - M_2 > 0.000020066721204449853370212. \quad (35)$$

Proof. This is a finite calculation over $\mathbb{Q}$, recorded in Appendix B. After clearing content, exact Euclidean algorithms prove that H is squarefree and coprime to P , and the Vincent–Akritas–Strzeboński continued-fraction method isolates every real root. The endpoints 0, 25, 49 and every other zero of a factor have $g(t) = 0$; therefore every positive maximum occurs at one of the isolated roots of H . Each factor is bounded on each rational isolating interval by exact Horner interval arithmetic. The logarithms are enclosed using Lemma 5.1. Comparing the resulting rational intervals gives (33)–(35). All 63 isolating intervals and their exact rational value bounds are included in the supplementary certificate. $\square$

By Lemma 6.1, Proposition 4.3, and (28),

$$\lim_{\substack{n \rightarrow \infty \\ 2D|n}} \frac{1}{n} \log(C_n I_{j,n}) = c + M_j = -\tau_j \quad (j = 1, 2). \quad (36)$$

Lemma 6.2 shows that $\tau_2 > \tau_1$, and the numerical bounds in Section 8 show that both are positive. Since the two errors are positive, they are nonzero for every admissible n .

7. Growth bound for the common coefficient

Put

$$Q_k(z) = P_k((z - 35)^2).$$

The coefficient of the logarithms in Proposition 4.3 is controlled by

$$a_{1,n} = [z^n] \frac{A_n(z)}{(70 - z)^{n+1}}. \quad (37)$$

We apply Cauchy's estimate on the circle $|z| = 26$.

Write $z = 26e^{i\theta}$ and $u = \cos \theta$. Define the integer polynomials

$$A_k^\circ(u) = |Q_k(26e^{i\theta})|^2 \quad (38)$$

and

$$B^\circ(u) = |26e^{i\theta}(70 - 26e^{i\theta})|^2 = 676(5576 - 3640u). \quad (39)$$

Set

$$F(u) = \frac{1}{2} \sum_{k=1}^{21} \alpha_k \log A_k^\circ(u) - \frac{1}{2} \log B^\circ(u), \quad -1 \leq u \leq 1, \quad (40)$$

and let $L = \max_{[-1,1]} F$.

Let $\mathcal{A}(u) = \prod_k A_k^\circ(u)$. The primitive integer numerator of $2DF'(u)$ is the primitive part, with positive content removed, of

$$B^\circ(u) \sum_{k=1}^{21} W_k(A_k^\circ)'(u) \frac{\mathcal{A}(u)}{A_k^\circ(u)} - D(B^\circ)'(u) \mathcal{A}(u). \quad (41)$$

Denote it by $J(u)$.

Lemma 7.1 (circle maximum). *Every A_k° is strictly positive on $[-1, 1]$. The polynomial J has degree 170 and factors as*

$$J(u) = (901u + 899)K(u), \quad (42)$$

where K has degree 169 and is strictly negative on $[-1, 1]$. Consequently F has its unique maximum at

$$u_0 = -\frac{899}{901}.$$

Exact outward logarithm bounds at u_0 give

$$4.537901075598406403746477 < L < 4.537901075598406404986681. \quad (43)$$

Proof. Map $u = 2v - 1$ from $[-1, 1]$ to $[0, 1]$. If $p(v) = \sum_{j=0}^n a_j v^j$, its degree- n Bernstein coefficients are

$$b_k = \sum_{j=0}^k a_j \frac{\binom{k}{j}}{\binom{n}{j}} \quad (0 \leq k \leq n). \quad (44)$$

Exact conversion by (44) gives positive Bernstein coefficients for every $A_k^\circ(2v-1)$ and negative Bernstein coefficients for $K(2v-1)$. Hence every circle factor is positive on $[-1, 1]$, and the sign of J is positive to the left of u_0 and negative to its right. The same calculation gives $J'(u_0) < 0$. The complete coefficient and sign certificate is described in Appendix B. Finally, Lemma 5.1 gives (43). $\square$

Proposition 7.2 (Cauchy growth bound). *Along the admissible values of n ,*

$$\limsup_{\substack{n \rightarrow \infty \\ 2D|n}} \frac{1}{n} \log |q_n| \leq c + L. \quad (45)$$

Proof. From (37) and Cauchy's coefficient estimate,

$$\begin{aligned} |a_{1,n}| &\leq \max_{|z|=26} \frac{|A_n(z)|}{26^n |70 - z|^{n+1}} \\ &\leq \frac{1}{44} \left(\max_{|z|=26} \frac{\prod_k |Q_k(z)|^{\alpha_k}}{|z(70 - z)|} \right)^n = \frac{1}{44} e^{nL}. \end{aligned}$$

Since $q_n = C_n a_{1,n}$, equation (28) gives (45). If a coefficient vanishes, the corresponding logarithm is interpreted as $-\infty$. By (36), both positive errors lie in $(0, 1)$ for every sufficiently large admissible n . If $q_n = 0$ at such an index, (24) would make the first error a nonzero integer, a contradiction. Thus q_n is eventually nonzero. $\square$

8. Certified constants and completion

The exact rational logarithm enclosure gives

$$1.300404222911437365598758 < c < 1.300404222911437366032151. \quad (46)$$

Combining the outward upper bounds for c , M_1 , and L gives

$$\tau_1 = -(c + M_1) > 1.422553272756972345019606, \quad (47)$$

$$\lambda := c + L < 5.838305298509843771018831. \quad (48)$$

The rational endpoints in these enclosures satisfy, by exact integer cross-multiplication,

$$\frac{\lambda}{\tau_1} < 4.104103101316511697013643 < \frac{513013}{125000} = 4.104104. \quad (49)$$

For $m \geq 1$, reindex the construction by $n_m = 2Dm$. Equations (36) and (45) then satisfy the hypotheses of Lemma 2.1, with every exponential constant multiplied by $2D$ and hence with the quotient in (49) unchanged. Apply the lemma to

$$\theta_1 = \log \frac{4}{3} = 2 \log 2 - \log 3, \quad \theta_2 = \log \frac{9}{8} = -3 \log 2 + 2 \log 3.$$

The coefficient transformation is unimodular. Explicitly, for arbitrary $h_1, h_2 \in \mathbb{Z}$, put

$$u = 2h_1 + 3h_2, \quad v = h_1 + 2h_2. \quad (50)$$

Then

$$u\theta_1 + v\theta_2 = h_1 \log 2 + h_2 \log 3.$$

Moreover, if $H = \max(|h_1|, |h_2|)$ and $H' = \max(|u|, |v|)$, then $H/5 \leq H' \leq 5H$. Choose ν strictly between the first and second quantities in (49). Lemma 2.1 gives

$$|h_0 + h_1 \log 2 + h_2 \log 3| > (H')^{-\nu} \geq (5H)^{-\nu}.$$

For all sufficiently large H , the last expression exceeds $H^{-4.104104}$. This proves Theorem 1.1.

Proof of Corollary 1.2. Taking $(h_1, h_2) = (0, q)$ and $h_0 = -p$ in Theorem 1.1 gives

$$\left| \log 3 - \frac{p}{q} \right| > q^{-5.104104}$$

for all sufficiently large q and every $p \in \mathbb{Z}$.

For $\xi = \log 3 / \log 2$, take $(h_1, h_2) = (-p, q)$. Whenever $|\xi - p/q| < 1$, one has $\max(|p|, q) \leq (1 + |\xi|)q$. The theorem therefore gives a positive constant C_ξ such that

$$\left| \xi - \frac{p}{q} \right| = \frac{|q \log 3 - p \log 2|}{q \log 2} > C_\xi q^{-5.104104}$$

for all sufficiently large q . For every $\delta > 0$, the right-hand side exceeds $q^{-(5.104104+\delta)}$ once q is sufficiently large. Thus the defining approximation inequality has only finitely many solutions for every exponent greater than 5.104104. Together with the first lower bound, this proves both assertions. $\square$

Appendix A. Factors and valuation profiles

In the third column, coefficients are ordered from the leading coefficient to the constant term. The historical ID records the column in the exploratory factor pool.

Table A.1: The twenty-one symmetrized factors and their local coefficient profiles.

k	ID	coefficients, high to low	τ_7	τ_5	τ_3	τ_2
1	0	[1, -49]	2	0	1	3
2	1	[1, -25]	0	2	1	4
3	2	[1, 0]	2	2	0	0
4	43	[38553, -19202708, 1088493350, -17941472500, 90075015625]	8	6	2	16
5	99	[463, -2770460, 172374650, -2731137500, 12867859375]	7	6	2	15
6	105	[1579, 4041716, -242861150, 3823592500, -18015003125]	8	5	1	16
7	124	[11, -38729, 834225, -4501875]	4	4	1	12
8	144	[16201, -64316693, 6078999290, -194217490250, 2344523978125, -9457876640625]	9	7	2	20
9	148	[3723, -14434700, 867361250, -13655687500, 64339296875]	7	7	0	16
10	159	[154073, -706525365, 69673778650, -2292537826250, 27833179828125, -110341894140625]	10	8	2	20
11	161	[13627, -789705, 11790625, -52521875]	5	5	1	12
12	163	[4259, -19924359, 1880315710, -60526208750, 754056559375, -3152625546875]	9	7	1	20
13	178	[8831, -33425731, 2301541270, -49661683750, 434933646875, -1351125234375]	8	7	2	20
14	189	[11881, -45475525, 3256944250, -74061846250, 685673078125, -2251875390625]	8	8	2	19
15	190	[31809, -138944253, 9336110826, -192597295450, 1574511273125, -4413675765625]	10	6	2	20
16	195	[85343, -320906467, 22689947350, -517088363750, 4799711546875, -15763127734375]	9	8	3	19
17	198	[21629, -86995629, 6252170386, -142667059850, 1329507230625, -4413675765625]	10	6	2	18
18	199	[359787, -1983981727, 196332531150, -6510548598750, 80617138984375, -331025682421875]	10	8	3	20
19	201	[14179, -60400263, 4236862910, -96560416750, 918765159375, -3152625546875]	9	7	3	20
20	202	[43951, -94114083, 9117300150, -301286483750, 3770282796875, -15763127734375]	9	8	2	19
21	208	[162521, -717601717, 70580396250, -2311361666250, 27833179828125, -110341894140625]	10	8	2	19

Appendix B. Computational certificate and reproducibility

This appendix records the implementation of the finite calculations used in Lemmas 6.2 and 7.1 and in the final comparison (49). The analytic and

arithmetic reductions are proved in the main text; the role of the software is limited to polynomial identities, real-root counts, rational interval inequalities, and exact sign decisions.

Appendix B.1. Origin and exact rationalization of the parameters

The active factor support arose from a discretized optimization of a semi-infinite linear program. The surviving exploratory output had the nonrigorous design value $5.1040765815\dots$. The submitted construction was then rationalized by imposing the circle point (10), prescribing a separation margin between the two real maxima, fixing the last fifteen weights with denominator 10^{12} , and solving (12)–(15) exactly for the first six weights. The exploratory value is not used in the proof. The supplement contains the surviving design output and a numerical-to-exact history so that this distinction can be audited.

Appendix B.2. Primary exact certificate

The program `verify_log3_bound_5_104104.py` uses Python integer and rational arithmetic together with SymPy’s exact polynomial algorithms [9]. Starting from the factor coefficients and the fifteen free weights displayed in Section 3, it

1. reconstructs the first six weights and checks their positivity and all six defining equations;
2. recomputes the twenty-one local coefficient profiles and all degree and content balances;
3. constructs the degree-85 polynomial H , proves squarefreeness and coprimality, and isolates its 63 roots in $[0, 49]$;
4. encloses every real critical value and proves the strict separation of M_1 and M_2 ;
5. constructs the degree-170 circle polynomial J , verifies $J = (901u + 899)K$, and proves the required Bernstein signs;
6. encloses L , c , λ , and τ_1 by rational intervals; and
7. proves (49) by integer cross-multiplication.

There is no floating-point branch or comparison, and the program refuses to run when Python assertions are disabled. Decimal output is diagnostic only. The exact denominator, weight numerators, root intervals, value bounds, and polynomial data are exported to `certificate_log3_5_104104.json`.

For identification of the reconstructed polynomials, the SHA-256 hashes of their primitive high-to-low coefficient lists are

H: 91e83aa85d3e09743bd8664736bd4edf764a52b945bd46592b3bf98fc69749c3,
J: 71a2327cc73fa1909a5e90933b438db0c547d903cdc6e31021f2e82a0995ce92.

Every logarithm comparison implements Lemma 5.1. A positive rational number is first reduced, by an exact power of two, to a number y satisfying $1 \leq y < 2$; the program then takes 24 positive-series terms, appends the explicit rational tail, and rounds outward to denominator 10^{18} .

Appendix B.3. Independent exact reproduction

The second program, `independent_exact_audit.py`, uses only Python’s standard library and imports neither the primary verifier nor a computer-algebra system. It separately implements rational Gaussian elimination, integer-polynomial arithmetic, logarithm enclosures, Bernstein conversion, and the final comparison. It reconstructs the coefficient lists and hashes of H and J . Exact Descartes transformations prove that each of the 63 exported intervals contains one root of H and that all 64 complementary gaps in $(0, 49)$ contain none. The JSON certificate is common input, but every theorem-facing entry used by this program is checked rather than assumed.

As a further cross-check, `audit_flint_crosscheck.py` passes the reconstructed integer polynomials to FLINT/Arb [10]. Rigorous complex-ball isolation independently confirms the 41/22 split of the roots of H , one root in every exported interval, and the absence of roots of K and of the circle factors in $(-1, 1)$. This third computation is logically redundant but guards against an implementation-specific root-isolation error.

Appendix B.4. Reproduction

The supplement’s README gives the complete commands and pinned dependencies. The primary run writes a fresh JSON certificate for bitwise comparison with the submitted one; the manuscript-consistency check and the two independent audits are then run separately. The archive supplies reference outputs and a SHA-256 manifest for every file.

Data and code availability

The manuscript source is supplied with the submission. The exact verifier, CAS-free independent reproduction, JSON certificate, expected outputs, file hashes, numerical-to-exact history, exploratory transcript, and independent FLINT/Arb root audit accompany this article as supplementary material. The primary verifier needs no network access and makes no floating-point proof decision.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT and Codex (OpenAI) for computational exploration, generation and checking of proof components and software, and drafting and editing the manuscript. After using these tools, the author reviewed and edited the resulting mathematical arguments, computations, code, and text as needed and takes full responsibility for the content of the article.

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